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//Program:Binary Search
//AIM:WAP to implement binary search operation using arrays
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#include <stdio.h>
int BinarySearch(int arr[],int n,int key);
int main(){
    int n,key;
    printf("enter the number of entries to the array\n");
    scanf("%d",&n);
    int arr[n];
    printf("enter the elements to the array\n");
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    printf("enter the element to be serached");
    scanf("%d",&key);
    int pos=BinarySearch(arr,n,key);
    if(pos==-1){
        printf("Element not found\n");
    }
    else{
        printf("element found at position %d",pos+1);
    }
    return 0;
}

int BinarySearch(int arr[],int n,int key){
    int low=0;
    int high=n-1;
    int mid;

    while(low<high){
        int mid=(low+high)/2;
        if(arr[mid]==key){
            return mid;
        }
        else if(arr[mid]<key)
            low=mid+1;
        else
            high=mid-1;
    }
    return -1;
}

```

